Layer 0: Wound Biology

Protease Cascade ODE: MMP-9 and HNE in Wound Fluid

Eton iGEM 2026, Dry Lab | Layer 0 of 17-Layer Biosensor Pipeline | `wound_biology.py`

Layer 0 predicts the **free protease concentrations in wound fluid at the time of swab sampling**. Its two key outputs, $[\text{MMP-9}]_{\text{free}}(t)$ and $[\text{HNE}]_{\text{free}}(t)$, are in-situ wound fluid concentrations that pass to Layer 1 (sample collection) via a swab dilution factor D_f.

The electrochemical device measures these concentrations *indirectly* via peptide substrate cleavage. MMP-9 and HNE each cleave a surface-immobilised peptide probe bearing a methylene-blue (MB) redox label; cleavage releases MB into solution, producing a signal-off decrease in SWV peak current proportional to enzymatic activity. The sensor therefore reports *activity*, not protein concentration directly. Layer 0 provides the free enzyme concentration that drives this activity; the kinetic link between $[\text{MMP-9}]_{\text{free}}$ and cleavage rate requires k_{cat} and K_m for the specific peptide substrate (Section V, Limitation 5).

The model tracks nine coupled species through a system of ordinary differential equations (Section IV), solved numerically with a stiff implicit integrator. Every parameter value must be grounded in a paper physically present in `papers/layer_00/`; values that are estimates rather than text-confirmed are explicitly labelled throughout.

I. Wound Fluid Composition

Biological rationale

Chronic wounds are characterised by a prolonged inflammatory phase sustained by persistent neutrophil and macrophage infiltration [1]. This drives elevated secretion of two serine and metalloprotease families that are the targets of this device. MMP-9 (gelatinase B, a zinc-dependent matrix metalloprotease) is secreted as an inactive zymogen (pro-MMP-9); activation requires proteolytic removal of the prodomain. HNE (human neutrophil elastase) is released already active from azurophil granules and is regulated primarily by α_1-antitrypsin (A1AT) and secondarily by secretory leukocyte protease inhibitor (SLPI).

Unit conversion

All species concentrations are held internally in mol/L. The conversion from the ng/mL units reported in ELISA literature is:

$$C\left[\frac{\text{mol}}{\text{L}}\right] = \frac{c\left[\frac{\text{ng}}{\text{mL}}\right] \times 10^{-6}\left[\frac{\text{g/L}}{\text{ng/mL}}\right]}{M_W\left[\frac{\text{g}}{\text{mol}}\right]} \tag{I.1}$$

For μg/mL values the prefactor is 10^{-3} instead of 10^{-6}. Applying Eq. (I.1) to active MMP-9 in chronic wound fluid (4800 ng/mL, $M_W = 92{,}000$ g/mol):

$$C_{\text{MMP-9}} = \frac{4800 \times 10^{-6}}{92{,}000} = 5.22 \times 10^{-8}\ \text{mol/L} = 52\ \text{nM}$$

Literature values

Ladwig et al. (2002) [2] used quantitative ELISA and gelatin zymography on wound fluid from 56 patients with chronic pressure ulcers. All values below are from paper text (p. 31):

- Pro-MMP-9: $8{,}100 \pm 3{,}000$ ng/mL; active MMP-9: $4{,}800 \pm 1{,}300$ ng/mL.
- Total MMP-9 = 12,900 ng/mL; pro-form fraction = 63%.
- MMP-9:TIMP-1 ratio inversely correlated with healing (text pp. 33–34): higher active MMP-9 and lower TIMP-1 in non-healing wounds.

Trengove et al. (1999) [1] used ELISA in venous leg ulcer (VLU) wound fluid. Values from text (p. 448):

- TIMP-1 in chronic VLU: median 0.8 μg/mL = 800 ng/mL ($n = 5$).
- HNE in non-healing VLU: median 1.4 μg/mL $\approx$ 47 nM (range $<$ 1–11.9 μg/mL; elevated in 6 of 13 chronic samples).
- MMP activity significantly lower in healing wounds ($p < 0.01$, text p. 449).

Rao et al. (1995) [7] examined A1AT integrity by western blot and elastase-inhibitory capacity (text pp. 572–575):

- A1AT intact and functional in all healing and acute wound samples.
- A1AT degraded and non-functional in 9 of 10 chronic wound samples.
- No quantitative ng/mL wound-fluid A1AT concentration is given in Rao 1995. The healing-state estimate $[A_0] \approx 1.92$ μM (serum $\approx$ 2 mg/mL at 10-fold dilution into wound fluid) is used; Section III shows the qualitative conclusion is insensitive to this estimate.

Nwomeh et al. (1999) [8] reported that TIMP-1 was "very low or not detected" in chronic ulcer fluid (text p. 193), consistent with Trengove 1999 [1] and confirming TIMP-1 suppression as a hallmark of the chronic wound environment. Values are expressed per μg protein and cannot be converted to ng/mL without the protein content data, so Trengove 1999 is used for quantitative TIMP-1.

Initial conditions

Table 1 gives the initial conditions for both wound states.

Species	Chronic wound	Healing wound	Confirmed?	Source
Total MMP-9	12,900 ng/mL (140 nM)	5,400 ng/mL (59 nM)	Chronic yes / Healing fig.	[2]
Pro-MMP-9 (63%)	8,100 ng/mL (88 nM)	3,400 ng/mL (37 nM)	Chronic text / Healing est.	[2]
Active MMP-9 (37%)	4,800 ng/mL (52 nM)	2,000 ng/mL (22 nM)	Chronic text / Healing fig.	[2]
TIMP-1	800 ng/mL (28 nM)	1,104 ng/mL (39 nM)	Chronic text / Healing none	[1]
HNE	47 nM (1.4 μg/mL), both states		Text	[1]
A1AT (functional)	$\approx$ 0 (degraded)	1.92 μM (est.)	Qualitative	[7]
SLPI	$\approx$ 200 ng/mL ($\approx$ 17 nM)		Not confirmed	est.

Table 1: Initial conditions for the Layer 0 ODE. Healing MMP-9 is read from Ladwig 2002 Figure 8 (approximate, not stated in text). Healing TIMP-1 has no paper source. A1AT wound-state status is qualitative from Rao 1995 [7] with an estimated concentration.

II. MMP-9 Inhibition by TIMP-1

TIMP-1 binds reversibly to the MMP-9 active site in non-covalent slow-tight inhibition. Denoting free MMP-9 as M, free TIMP-1 as T, and their complex as C_1, the rate equations are:

$$\frac{d[M]}{dt} = -k_{\text{on}}^{T}\,[M][T] + k_{\text{off}}^{T}\,[C_1] \tag{II.1}$$

$$\frac{d[T]}{dt} = -k_{\text{on}}^{T}\,[M][T] + k_{\text{off}}^{T}\,[C_1] \tag{II.2}$$

$$\frac{d[C_1]}{dt} = +k_{\text{on}}^{T}\,[M][T] - k_{\text{off}}^{T}\,[C_1] \tag{II.3}$$

Olson et al. (1997) [4] reported kinetic constants from solution-phase enzyme inhibition assays (Table IV, 82 kDa active MMP-9):

$$k_{\text{on}}^{T} = 2.48 \pm 0.47 \times 10^{5}\ \text{M}^{-1}\text{s}^{-1}$$
$$k_{\text{off}}^{T} = 2.1 \pm 1.1 \times 10^{-3}\ \text{s}^{-1}$$

The equilibrium inhibition constant K_i is stated directly in Olson 1997 Table IV as $K_i = 8.5 \pm 2.9$ nM; the ratio confirms this:

$$K_i = \frac{k_{\text{off}}^{T}}{k_{\text{on}}^{T}} = \frac{2.1 \times 10^{-3}}{2.48 \times 10^{5}} = 8.47\ \text{nM} \approx 8.5\ \text{nM} \tag{II.4}$$

Enzyme inhibition values are used in preference to the SPR values in Olson 1997 Tables I and II because solution-phase kinetics are more representative of wound-fluid conditions; k_{on}^{T} from enzyme inhibition is 2.5–7$\times$ higher than from SPR (Olson 1997, p. 29981).

Equilibrium free MMP-9

The TIMP-1/MMP-9 equilibration timescale is:

$$\tau = \frac{1}{k_{\text{on}}^{T}\,[T_0]} = \frac{1}{2.48 \times 10^{5} \times 28 \times 10^{-9}} \approx 144 \text{ s} \approx 2.4 \text{ min}$$

This is short relative to the 30-min measurement window, so the system rapidly reaches an approximate binding equilibrium. Setting $d[C_1]/dt = 0$ in Eq. (II.3) gives:

$$K_i = \frac{[M][T]}{[C_1]} \tag{II.5}$$

Conservation of each species pool requires:

$$[M_0] = [M] + [C_1] \tag{II.6}$$
$$[T_0] = [T] + [C_1] \tag{II.7}$$

From Eq. (II.6): $[C_1] = [M_0] - [M]$. Substituting into Eq. (II.7): $[T] = [T_0] - [M_0] + [M]$. Substituting both into Eq. (II.5) and writing $x = [M]_{\text{free}}$:

$$K_i\,(M_0 - x) = x\,(T_0 - M_0 + x)$$

Expanding and collecting terms:

$$x^2 + x\,(T_0 - M_0 + K_i) - K_i\,M_0 = 0 \tag{II.8}$$

Eq. (II.8) is a quadratic in x. Taking the positive root (the only physically meaningful solution, since $[M]_{\text{free}} \geq 0$):

$$[M]_{\text{free}} = \frac{-(T_0 - M_0 + K_i) + \sqrt{(T_0 - M_0 + K_i)^2 + 4\,K_i\,M_0}}{2} \tag{II.9}$$

Chronic wound ($M_0 = 52$ nM, $T_0 = 28$ nM, $K_i = 8.5$ nM):

$$b = T_0 - M_0 + K_i = 28 - 52 + 8.5 = -15.5 \text{ nM}$$
$$[M]_{\text{free}} = \frac{15.5 + \sqrt{(-15.5)^2 + 4 \times 8.5 \times 52}}{2}$$
$$= \frac{15.5 + \sqrt{240.3 + 1768}}{2} = \frac{15.5 + 44.8}{2} \approx \mathbf{30.2} \text{ nM}$$

Healing wound ($M_0 = 22$ nM, $T_0 = 39$ nM, $K_i = 8.5$ nM):

$$b = T_0 - M_0 + K_i = 39 - 22 + 8.5 = +25.5 \text{ nM}$$
$$[M]_{\text{free}} = \frac{-25.5 + \sqrt{(25.5)^2 + 4 \times 8.5 \times 22}}{2}$$
$$= \frac{-25.5 + \sqrt{650.3 + 748}}{2} = \frac{-25.5 + 37.4}{2} \approx \mathbf{5.9} \text{ nM}$$

These analytical equilibrium values agree closely with the full ODE result at $t = 30$ min (31.9 nM chronic, 6.2 nM healing; Table 2). The small difference reflects ongoing pro-MMP-9 activation driving the system slightly away from the pure TIMP-1 equilibrium.

Sensitivity to K_i
Changing K_i from 8.5 nM to 1 nM in Eq. (II.9) shifts the chronic result from 30.2 nM to 26.5 nM (a 12% decrease). The healing result is more sensitive: at $K_i = 1$ nM it falls from 5.9 nM to 1.4 nM, because the TIMP-1 excess (17 nM) is comparable to K_i. The reported uncertainty (±2.9 nM) is within this regime; the healing estimate should be treated as order-of-magnitude.

The following assumptions underlie this section:

1. The equilibrium approximation is valid because $\tau \approx 2.4$ min is short relative to the 30-min read window and the pro-MMP-9 activation timescale.
2. Solution-phase enzyme inhibition constants (Olson 1997 [4] Table IV) are used in preference to SPR values, as the wound fluid environment is closer to solution phase.
3. Pro-MMP-9 activation (Section IV) is negligible with current placeholder kinetics; the active MMP-9 initial condition is approximately constant over the measurement window.

III. HNE Inhibition

A1AT/HNE near-irreversible binding
A1AT inhibits HNE via serpin mechanism: a covalent acyl-enzyme intermediate forms and the complex is effectively irreversible ($k_{\text{off}}^{A} \approx 0$). The rate of free HNE depletion is:

$$\frac{d[H]}{dt} = -k_{\text{on}}^{A}\,[A][H] \tag{III.1}$$

In healing wounds where $[A_0] \gg [H_0]$ (see below), the free A1AT concentration remains approximately constant throughout the reaction. This pseudo-first-order approximation reduces Eq. (III.1) to:

$$\frac{d[H]}{dt} \approx -k^{*}\,[H], \qquad k^{*} = k_{\text{on}}^{A}\,[A_0] \tag{III.2}$$

Eq. (III.2) is a first-order linear ODE with exponential solution:

$$[H](t) = [H_0]\,\exp\!\big(-k_{\text{on}}^{A}\,[A_0]\,t\big) \tag{III.3}$$

The half-life, at which $[H] = [H_0]/2$, follows from setting $\exp(-k^{*}t_{1/2}) = 1/2$:

$$t_{1/2} = \frac{\ln 2}{k_{\text{on}}^{A} \cdot [A_0]} \tag{III.4}$$

Beatty, Bieth & Travis (1980) [5] confirmed $k_{\text{on}}^{A} = 6.5 \pm 4.0 \times 10^{7}$ $\text{M}^{-1}\text{s}^{-1}$ from direct kinetic measurement (Table 1, human leukocyte elastase vs. native α_1-PI). With the healing-wound estimate $[A_0] = 1.92$ μM:

$$t_{1/2} = \frac{0.693}{6.5 \times 10^{7} \times 1.92 \times 10^{-6}} = \frac{0.693}{124.8} \approx 5.5 \text{ ms} \tag{III.5}$$

Sensitivity to $[A_0]$. At $[A_0]$ ten-fold lower (0.192 μM):

$$t_{1/2} = \frac{0.693}{6.5 \times 10^{7} \times 0.192 \times 10^{-6}} \approx 55 \text{ ms} \tag{III.6}$$

Even at this lower bound, $t_{1/2} = 55$ ms remains three orders of magnitude below the 5-min practical minimum measurement window. The healing-wound conclusion, $[\text{HNE}]_{\text{free}} \approx 0$, is therefore insensitive to the [A1AT] estimate over a 10-fold range.

Validity check. In healing wounds $[A_0] = 1.92$ μM $\gg [H_0] = 47$ nM, confirming the pseudo-first-order approximation. After complete HNE absorption: $[A]_{\text{free}} \approx 1.92$ μM − 47 nM ≈ 1.87 μM (confirmed by ODE, Table 2).

A1AT wound-state dependence

Rao et al. (1995) [7] showed that functional A1AT cannot be assumed present across wound types:

- **Healing and acute wounds**: A1AT intact and functional in all samples. Eq. (III.5) applies; $[\text{HNE}]_{\text{free}} = 0$ within milliseconds.
- **Chronic wounds**: A1AT degraded and non-functional in 9 of 10 samples. With functional $[A_0] \approx 0$, Eq. (III.1) gives $d[H]/dt \approx 0$: HNE persists in wound fluid at near its initial concentration (moderated only by SLPI).

This A1AT wound-state switch provides a second discriminating axis: healing wounds have $[\text{HNE}]_{\text{free}} \approx 0$ while chronic wounds (where A1AT is degraded) retain elevated free HNE. However, HNE elevation itself was detected in only 6 of 13 chronic VLU samples in Trengove 1999 [1], meaning ∼50% of chronic cases are HNE-low regardless of A1AT status. HNE is therefore not a reliable standalone discriminator (see also Section V).

SLPI secondary inhibition in chronic wounds

In chronic wounds where A1AT is non-functional, SLPI provides the residual HNE inhibition. Ashcroft et al. (2000) [6] confirmed SLPI's role in wound healing using knockout mice but did not report a quantitative wound-fluid SLPI concentration. The model uses placeholder kinetics ($k^S_{\text{on}} = 10^5$ $\text{M}^{-1}\text{s}^{-1}$, $k^S_{\text{off}} = 10^{-4}$ s^{-1}, $[S_0] \approx 200$ ng/mL ≈ 17 nM); with these values the ODE gives $[\text{HNE}]_{\text{free}} \approx 30$ nM at $t = 30$ min in chronic wounds. This prediction is sensitive to the placeholder SLPI kinetics.

The following assumptions underlie this section:

1. Pseudo-first-order kinetics are valid for healing wounds because $[A_0] = 1.92\ \mu\text{M} \gg [H_0] = 47$ nM.
2. $[A_0] = 1.92\ \mu$M is a serum/10 estimate; the half-life conclusion is insensitive over a 10-fold range (Eq. (III.6)).
3. k^A_{on} was measured at 25°C, pH 8.0 in Beatty 1980 [5]; applied here at wound pH 7.4 as the best available value.
4. SLPI concentration and kinetics are unconfirmed placeholders; the chronic HNE prediction is not considered quantitatively reliable.

IV. Full ODE System

Species and notation

Nine coupled species track the wound fluid protease cascade. All concentrations in mol/L:

P	pro-MMP-9	C_1	MMP-9:TIMP-1 complex
M	free (active) MMP-9	H	free HNE
T	free TIMP-1	A	free A1AT
C_2	HNE:A1AT complex	S	free SLPI
C_3	HNE:SLPI complex		

Governing equations

Pro-MMP-9 is activated by MMP-3 and plasmin in excess, so the activation is pseudo-first-order in P with effective rate k^{**}:

$$\dot{P} = -k^{**}\,P \tag{IV.1}$$

$$\dot{M} = +k^{**}\,P - k^T_{\text{on}} MT + k^T_{\text{off}} C_1 \tag{IV.2}$$

$$\dot{T} = -k^T_{\text{on}} MT + k^T_{\text{off}} C_1 \tag{IV.3}$$

$$\dot{C}_1 = +k^T_{\text{on}} MT - k^T_{\text{off}} C_1 \tag{IV.4}$$

$$\dot{H} = -k^A_{\text{on}} HA - k^S_{\text{on}} HS + k^S_{\text{off}} C_3 \tag{IV.5}$$

$$\dot{A} = -k^A_{\text{on}} HA \tag{IV.6}$$

$$\dot{C}_2 = +k^A_{\text{on}} HA \tag{IV.7}$$

$$\dot{S} = -k^S_{\text{on}} HS + k^S_{\text{off}} C_3 \tag{IV.8}$$

$$\dot{C}_3 = +k^S_{\text{on}} HS - k^S_{\text{off}} C_3 \tag{IV.9}$$

where $k^{**} = (k_{\text{cat}}/K_m)_{\text{MMP-3}}\,[\text{MMP-3}] + (k_{\text{cat}}/K_m)_{\text{plasmin}}\,[\text{plasmin}]$ is the total first-order activation rate; k^T_{on}, k^T_{off} from Olson 1997 [4]; k^A_{on} from Beatty 1980 [5]; k^S_{on}, k^S_{off} are placeholders.

Mass conservation

Adding the appropriate equations confirms that each species pool is separately conserved:

$$\begin{aligned} \frac{d}{dt}(P + M + C_1) &= 0 \qquad \text{(total MMP-9)} \\ \frac{d}{dt}(T + C_1) &= 0 \qquad \text{(total TIMP-1)} \\ \frac{d}{dt}(H + C_2 + C_3) &= 0 \qquad \text{(total HNE)} \\ \frac{d}{dt}(A + C_2) &= 0 \qquad \text{(total A1AT)} \\ \frac{d}{dt}(S + C_3) &= 0 \qquad \text{(total SLPI)} \end{aligned}$$

These identities provide a numerical check: deviations indicate integration error. The Radau solver maintains conservation to within rtol $= 10^{-10}$, atol $= 10^{-15}$ tolerances.

Stiff integration

The system spans seven orders of magnitude in characteristic timescale:

- A1AT/HNE reaction: $t_{1/2} \approx 5.5$ ms (fastest)
- TIMP-1/MMP-9 equilibration: $\tau \approx 2.4$ min (intermediate)
- Pro-MMP-9 activation: $\sim$120 min (slowest; negligible with placeholders)

The timescale ratio $\sim 10^7$ makes the system stiff: explicit solvers would require $\Delta t \lesssim 1$ ms, making a 120-min integration prohibitively expensive. The Radau implicit solver (`scipy.integrate.solve_ivp`, method `Radau`) is used throughout.

V. Model Output and Wound-State Discrimination

Figures 1 and 2 show the ODE trajectories.

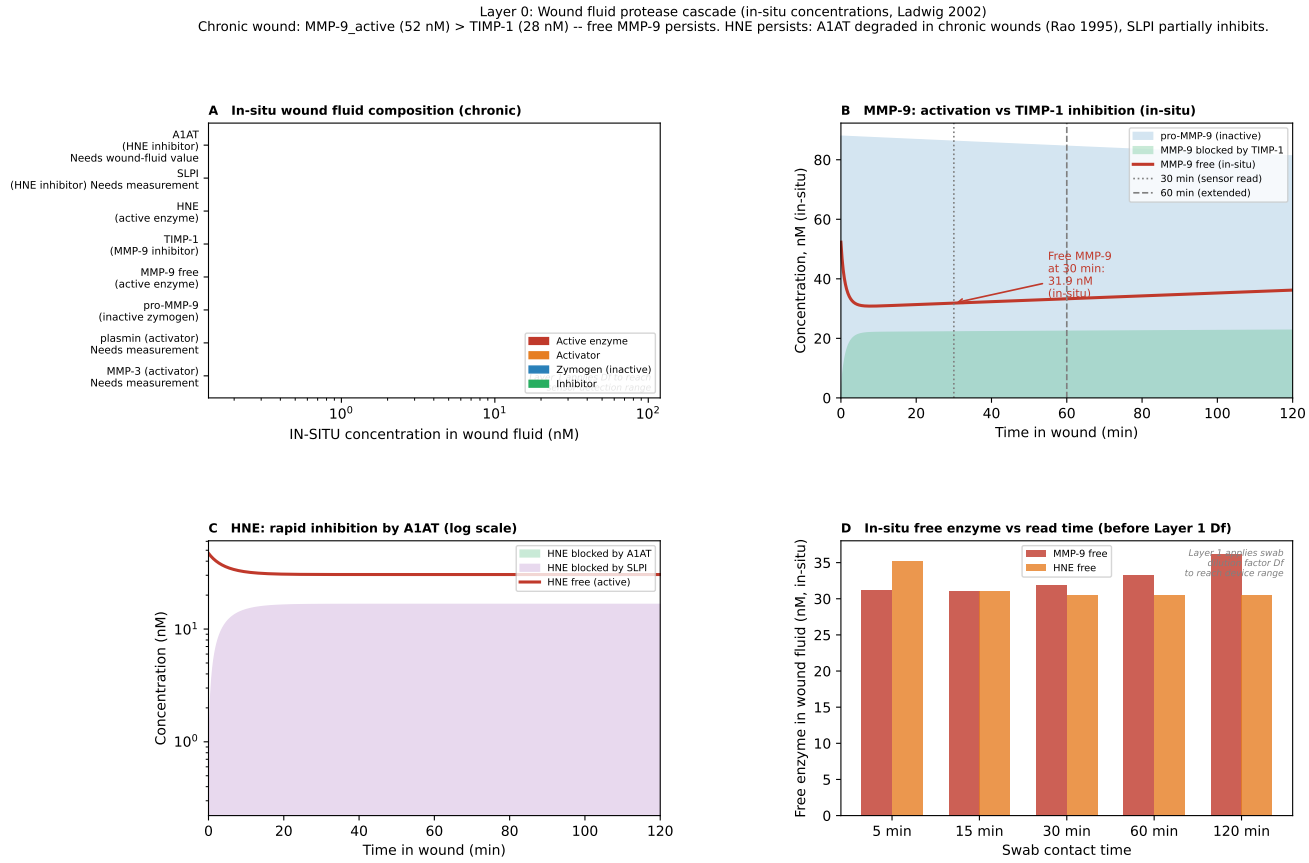


Figure 1: Layer 0 ODE output for the chronic wound state over 120 min. Top left: in-situ initial concentrations. Top right: MMP-9 activation vs. TIMP-1 inhibition. Bottom left: HNE in both wound states (log scale; note 5.5 ms half-life in healing). Bottom right: free enzyme at read times.

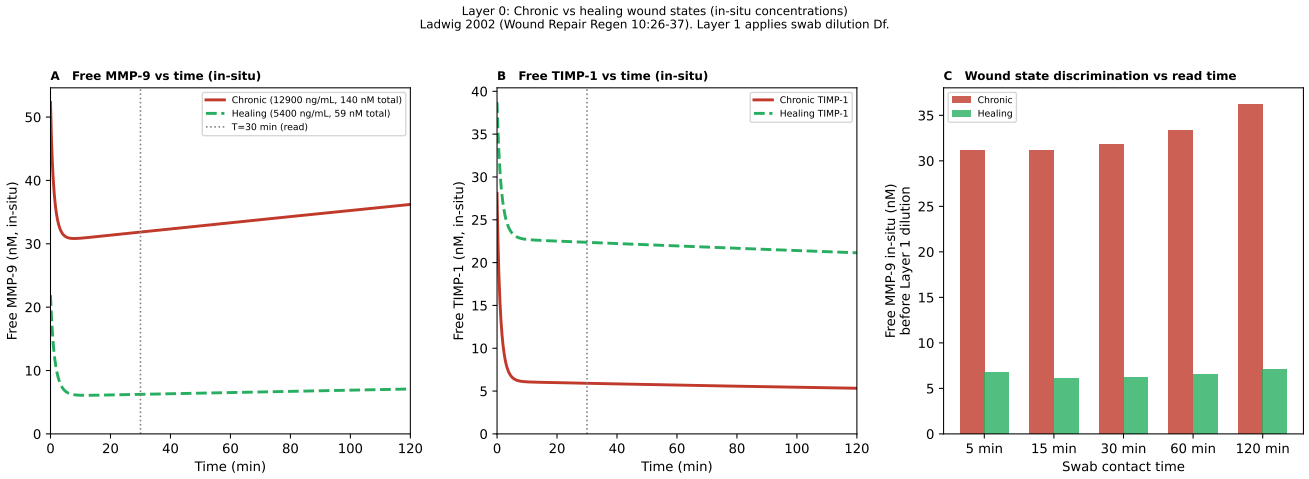


Figure 2: Chronic vs. healing wound states. Left: free MMP-9 over time. Centre: free TIMP-1. Right: MMP-9 discrimination at read times.

ODE results at $t = 30$ min

Species	Chronic wound	Healing wound	C/H ratio
Free MMP-9	31.9 nM	6.2 nM	5.1×
Free TIMP-1	5.9 nM	22.4 nM	–
Free HNE	30.5 nM †	0.00 pM	–
Free A1AT	0.00 μM	1.88 μM	–

Table 2: ODE output at $t = 30$ min. †A1AT degraded in 9/10 chronic samples (Rao 1995 [7]); HNE elevated in only 50% of chronic VLU (Trengove 1999 [1]). Chronic HNE prediction is sensitive to placeholder SLPI kinetics. Healing values depend on unconfirmed initial conditions.

MMP-9 wound-state discrimination

The 5.1× discrimination ratio arises from the sign of the MMP-9 vs. TIMP-1 excess, as made explicit by Eq. (II.9):

- **Chronic wound**: MMP-9 (52 nM) exceeds TIMP-1 (28 nM) by 24 nM. With $K_i = 8.5$ nM $\ll M_0, T_0$, TIMP-1 is nearly fully bound and free MMP-9 persists at ≈ 32 nM.
- **Healing wound** (estimated): TIMP-1 (39 nM) exceeds MMP-9 (22 nM) by 17 nM. TIMP-1 excess suppresses free MMP-9 to ≈ 6 nM.

The mechanism is structurally robust: for $K_i \ll M_0, T_0$, Eq. (II.9) gives $[M]_{\text{free}} \approx M_0 - T_0$ to first order (the excess simply remains free). Any wound state where active MMP-9 exceeds TIMP-1 will show elevated free MMP-9, independent of exact parameter values.

HNE wound-state discrimination

The A1AT wound-state switch (Rao 1995 [7], Section III) produces a strong binary in HNE: healing wounds have $[\text{HNE}]_{\text{free}} = 0.00$ pM, while chronic wounds with degraded A1AT show $[\text{HNE}]_{\text{free}} \approx 30$ nM. However, elevated HNE was observed in only 6/13 (50%) of chronic VLU samples (Trengove 1999 [1]); the remaining 50% of chronic wounds are HNE-low regardless of A1AT status. HNE is not a reliable primary discriminant; MMP-9 (5× discrimination, grounded in confirmed kinetics) is the preferred target. The second electrode targeting HNE serves as an internal reference rather than a primary channel.

Cleavage kinetics and sensor signal

The sensor signal is the fractional SWV current decrease:

$$\frac{\Delta I}{I_0} = \frac{I_0 - I(t)}{I_0} = f_{\text{cleaved}}(t) \tag{V.1}$$

where I_0 is the pre-incubation baseline current (all MB tethered) and $I(t)$ is the current at assay time t after wound

fluid exposure. Both measurements are made in the same buffer, so any Ca^{2+}-induced change in MB-electrode electron transfer rate applies equally to I_0 and $I(t)$ and cancels in the ratio (Layer 7).

The cleavage velocity per unit electrode area at the start of the assay is:

$$v = k_{\text{cat}} \cdot [\text{MMP-9}]_{\text{free}} \cdot \frac{\Gamma_p}{K_m^{\text{eff}} + \Gamma_p} \tag{V.2}$$

where Γ_p is the probe surface density (from Layer 7) and K_m^{eff} is the apparent Michaelis constant for surface-immobilised substrate. In the linear regime ($\Gamma_p \ll K_m^{\text{eff}}$):

$$v \approx \frac{k_{\text{cat}}}{K_m} \cdot [\text{MMP-9}]_{\text{free}} \cdot \Gamma_p \tag{V.3}$$

The wound-state discrimination ratio for sensor signal equals the ratio of cleavage velocities. In the linear regime, k_{cat}, K_m, and Γ_p cancel:

$$\frac{v_{\text{chronic}}}{v_{\text{healing}}} = \frac{[\text{MMP-9}]_{\text{free,chronic}}}{[\text{MMP-9}]_{\text{free,healing}}} \approx \frac{30\text{ nM}}{6\text{ nM}} \approx 5\times \tag{V.4}$$

If TIMP-1 inhibition is competitive with the peptide substrate, the apparent inhibition constant is modified by substrate occupancy:

$$K_{i,\text{app}} = K_i\left(1 + \frac{[S]}{K_m}\right) \tag{V.5}$$

where $[S]/K_m$ is the fractional saturation of the enzyme by substrate. In the linear regime ($[S] \ll K_m$), $K_{i,\text{app}} \approx K_i = 8.5$ nM and the ODE equilibrium is self-consistent. At higher substrate occupancy, TIMP-1 inhibition weakens ($K_{i,\text{app}} > K_i$), which would increase free MMP-9 in healing wounds relative to the ODE prediction and potentially reduce the discrimination ratio below 5×.

Kridel et al. (2001) [9] (Table II) measured kinetic parameters for MMP-9 cleaving peptide A11 (SGKIPRTLTA), the device substrate sequence (50 mM Tris, pH 7.5, 10 mM $CaCl_2$, 37°C):

$$k_{\text{cat}} = 86.4\text{ s}^{-1}, \quad K_m = 1.3\text{ mM}, \quad \frac{k_{\text{cat}}}{K_m} = 67{,}500\text{ M}^{-1}\text{s}^{-1} \tag{V.6}$$

The substrate competition correction at the electrode surface is discussed in Limitation 5.

Layer 1 dilution

The required dilution factor for the device entry concentration to fall within the enzyme activity detection window is established in Layer 1 using k_{cat}, K_m, and the signal sensitivity (LOD in terms of $\Delta I/I_0$). The solution-phase kinetic constants ($k_{\text{cat}} = 86.4$ s^{-1}, $K_m = 1.3$ mM) are now confirmed from Kridel 2001 [9]. The remaining blocker is the surface effective rate constant $k_{\text{eff,surf}}$, which must be measured on the fabricated electrode (Layer 9) before the LOD and dilution factor can be specified quantitatively. The in-situ free MMP-9 concentrations from Table 2 (30 nM chronic, 6 nM healing) are the inputs to the Layer 1 calculation.

Limitations of this model

Each limitation below states what is simplified, explains why the simplification does not invalidate the model's primary output (the MMP-9 wound-state discrimination ratio), and identifies what experiment would resolve it.

1. **Healing wound initial conditions are not text-confirmed.** The chronic wound values are the primary output of this model: active MMP-9 = 52 nM and TIMP-1 = 28 nM are both text-confirmed from Ladwig 2002 (p. 31) [2], and all kinetic parameters (K_i, k_{on}^T, k_{off}^T) are text-confirmed from Olson 1997 Table IV [4]. The healing MMP-9 estimate (5,400 ng/mL) is read from Ladwig 2002 Figure 8; the healing TIMP-1 (1,104 ng/mL) has no paper source. These estimates are used only to compute a comparison point, not to determine the chronic wound output. The model's structural conclusion, that any wound state in which active MMP-9 exceeds TIMP-1 will yield elevated free MMP-9 (from the sign of Eq. (II.9)), is independent of the healing estimates. The 5× ratio should be treated as a working figure. Resolution requires a joint VLU dataset reporting both MMP-9 and TIMP-1 from the same patients at chronic and healing time points.

2. **SLPI kinetics are unconfirmed placeholders and do not affect MMP-9.** The SLPI sub-system (Eqs. IV.8–IV.9) is completely decoupled from the MMP-9/TIMP-1 sub-system: SLPI does not bind MMP-9 or TIMP-1 and cannot alter the MMP-9 partition calculated in Section II. SLPI uncertainty propagates exclusively to the HNE secondary channel, which is already acknowledged as a limited discriminant (Limitation 4 below). The SLPI concentration and kinetics ($k_{\text{on}}^S = 10^5$ M^{-1}s^{-1}, $k_{\text{off}}^S = 10^{-4}$ s^{-1}) are order-of-magnitude placeholders; Ashcroft 2000 [6] confirms SLPI's role in wound healing from knockout mice but provides no quantitative wound-fluid SLPI concentration. Wound-fluid SLPI ELISA would resolve this for the HNE channel.

3. **Pro-MMP-9 activation is negligible at current parameter values.** The active MMP-9 initial condition (52 nM, chronic; 22 nM, healing) is taken directly from Ladwig 2002 [2] and does not depend on the activation sub-model. The activation kinetics (via MMP-3 and plasmin; Eq. IV.1) use order-of-magnitude placeholders, and with these values less than 1% of pro-MMP-9 converts in 120 min. This means the active MMP-9 pool is effectively constant over the 30-minute read window, consistent with using the Ladwig ELISA snapshot as the initial condition. These activation parameters become important only if wound-fluid MMP-3 or plasmin concentrations prove substantially higher than assumed, which would require a separate measurement.

4. **HNE elevation is inconsistent across chronic wound samples.** Elevated free HNE was detected in only 6 of 13 chronic VLU samples in Trengove 1999 [1]: 50% of chronic wounds are HNE-low regardless of A1AT status. The HNE channel therefore has limited diagnostic utility and is not the primary discriminant. This limitation does not affect MMP-9, which shows elevated free concentration in all chronic wound samples by the TIMP-1/MMP-9 excess mechanism (Section II). The HNE electrode is treated as a secondary reference channel, not a primary diagnostic output.

5. **k_{cat} and K_m for SGKIPRTLTA are now confirmed from Kridel et al. (2001) [9]; the substrate competition correction at the electrode surface must be quantified.**

   Kridel et al. (2001) [9] Table II reports solution-phase kinetics for MMP-9 cleaving peptide A11 (SGKIPRTLTA):

   $$k_{\text{cat}} = 86.4 \text{ s}^{-1}, \quad K_m = 1.3 \text{ mM}, \quad k_{\text{cat}}/K_m = 67{,}500 \text{ M}^{-1}\text{s}^{-1}$$

   measured by fluorescamine incorporation assay at pH 7.5, 10 mM $CaCl_2$, 37°C. These are solution-phase values; surface immobilisation systematically reduces k_{cat}/K_m (Layer 9), so they serve as an upper bound for the surface effective rate constant $k_{\text{eff,surf}}$.

   Wound fluid equilibrium. The Layer 0 ODE models the TIMP-1/MMP-9 partition in *wound fluid*, not at the electrode. TIMP-1 equilibrates with MMP-9 in the wound fluid bulk, where the relevant competing substrates are endogenous (collagen, gelatin) at concentrations that do not enter the ODE. The electrode peptide at $\Gamma_p \sim 3$ pmol cm^{-2} represents sub-nanomolar total moles in the assay volume and is a negligible perturbation on the wound fluid equilibrium. The ODE $K_i = 8.5$ nM (Olson 1997 [4]) therefore remains valid for the wound fluid partition.

   Justification for sequential treatment. A fully coupled competitive-inhibition scheme would require solving TIMP-1 inhibition and surface Michaelis–Menten kinetics simultaneously, since TIMP-1 and the surface peptide compete for the MMP-9 active site. The sequential approach used here, in which the wound fluid ODE is solved first to obtain $[\text{MMP-9}]_{\text{free}}$ and that value is then used as a constant input to the surface MM velocity, is valid under two conditions that are both satisfied. First, the wound fluid bulk contains no surface peptide substrate ($[S]_{\text{bulk}} \approx 0$), so the ODE inhibition equilibrium uses $K_i = 8.5$ nM without any substrate competition term, and the result is internally self-consistent. Second, the $[\text{MMP-9}]_{\text{free}}$ obtained from the wound fluid ODE is a *conservative lower bound* on electrode-accessible enzyme: at the electrode surface $K_{i,\text{app}} \approx 33$ nM (Eq. V.9), meaning TIMP-1 inhibition is weaker at the electrode than in the wound fluid. The sequential treatment therefore under-predicts, not over-predicts, the enzyme available for surface cleavage, and the discrimination ratio of 5× is conservatively estimated.

   Substrate depletion is negligible. The total surface-immobilised SGKIPRTLTA on the electrode is $\Gamma_p \times A \approx 5$ pmol cm^{-2} $\times$ 0.002 cm^2 = 10 fmol. The bulk MMP-9 in the assay volume is $\sim$0.1–0.2 pmol (at 5–10 nM in $\sim$20 μL). Even if all surface substrate were cleaved simultaneously, that represents $< 0.1\%$ of the total MMP-9 pool. Substrate depletion at this level cannot perturb the TIMP-1/MMP-9 binding equilibrium ($K_i = 8.5$ nM, Olson 1997 [4]), which requires a $\gtrsim$10% perturbation in free MMP-9 to shift the partition measurably. Consequently, $[\text{MMP-9}]_{\text{free}}$ from the wound fluid ODE remains accurate throughout the Layer 9 assay window, and no TIMP-1 competitive inhibition correction term is required in the Layer 9 k_{eff} expression.

   Electrode surface correction. At the electrode surface the effective local substrate concentration depends on probe density and probe-layer thickness. For $\Gamma_p = 3$ pmol cm^{-2} and effective layer thickness $d_{\text{eff}} \approx 8$ nm

(Layer 8):

$$[S]_{\text{eff}} = \frac{\Gamma_p}{d_{\text{eff}}} = \frac{3 \times 10^{-12}\ \text{mol cm}^{-2}}{8 \times 10^{-7}\ \text{cm}} \approx 3.75\ \text{mM} \tag{V.7}$$

Since $[S]_{\text{eff}}/K_m \approx 3.75/1.3 \approx 2.9$, the electrode surface is not in the linear regime. The Michaelis–Menten velocity is:

$$v = k_{\text{cat}} \cdot [\text{MMP-9}]_{\text{free}} \cdot \frac{\Gamma_p}{K_m + [S]_{\text{eff}}} \tag{V.8}$$

The wound-state discrimination ratio $v_{\text{chronic}}/v_{\text{healing}}$ still reduces to $[\text{MMP-9}]_{\text{chronic}}/[\text{MMP-9}]_{\text{healing}} \approx 5\times$ because K_m and $[S]_{\text{eff}}$ cancel identically in the ratio. The non-linear regime therefore does not reduce the discrimination ratio but does affect the absolute signal magnitude and the apparent TIMP-1 inhibition. If TIMP-1 inhibition is competitive with the surface peptide, the apparent inhibition constant at the electrode is:

$$K_{i,\text{app}} = K_i\left(1 + \frac{[S]_{\text{eff}}}{K_m}\right) \approx 8.5 \times (1 + 2.9) \approx 33\ \text{nM} \tag{V.9}$$

This weakens apparent TIMP-1 inhibition at the electrode surface, increasing the fraction of MMP-9 available for cleavage relative to the wound fluid ODE prediction. The surface k_{cat}/K_m must be measured directly on the fabricated electrode (Layer 9) before the absolute signal magnitude can be specified.

For HNE: A1AT inhibition is a serpin acyl-enzyme mechanism, not competitive inhibition. There is no $K_{i,\text{app}}$ correction. The half-life conclusion is insensitive to a 10-fold change in k_{on}^A (Eq. III.6). The k_{cat} and K_m for HNE cleaving the HNE peptide substrate are not yet confirmed from any paper in `papers/layer_00/` and remain an outstanding parameter for the HNE channel.

References

All papers in `electrochemical_device/papers/layer_00/`.

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[2] Ladwig GP, Robson MC, Liu R, Kuhn MA, Muir DF, Schultz GS (2002). Ratios of activated matrix metalloproteinase-9 to tissue inhibitor of metalloproteinase-1 in wound fluids are inversely correlated with healing of pressure ulcers. *Wound Repair Regen* 10(1):26–37. **ladwig2002.pdf** – chronic MMP-9 and TIMP-1 from text (p. 31) confirmed; healing values from Figure 8 only (approximate).

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[6] Ashcroft GS, Lei K, Jin W, Longenecker G, Kulkarni AB, Greenwell-Wild T, Hale-Donze H, McGrady G, Song XY, Wahl SM (2000). Secretory leukocyte protease inhibitor mediates non-redundant functions necessary for normal wound healing. *Nat Med* 6(10):1147–1153. **ashcroft2000.pdf** – SLPI role confirmed in knockout mouse model; no quantitative wound-fluid SLPI concentration reported.

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[9] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276(23):20572–20578. DOI: 10.1074/jbc.M100900200. **kridel2001.pdf** – $k_{\text{cat}} = 86.4\ \text{s}^{-1}$, $K_m = 1.3$ mM, $k_{\text{cat}}/K_m = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ for peptide A11 (SGKIPRTLTA) by MMP-9 confirmed from Table II. Measured by fluorescamine assay, pH 7.5, 10 mM $CaCl_2$, 37°C.